CarePath AI | Team Roles and Agentic Architecture

Shareable summary of the role allocation and build plan discussed in chat | UWE Responsible AI Hackathon, 24-25 October 2026

Team structure: Three active technical owners and one lighter research, evaluation and presentation role. The objective is to demonstrate one trustworthy patient journey from uploading a synthetic blood test to verifying deletion.

1. Team responsibilities

Member 1 | Technical lead: orchestration, privacy and security

- Implement OCR integration and Microsoft Presidio text/image redaction.
- Develop the bounded LangGraph workflow, consent and user approval gates.
- Implement safe session handling and deletion; own integration and safety tests.

Deliverable: Working orchestration, redacted reports, approval controls and verified deletion.

Member 2 | Explainability and RAG engineer

- Build retrieval over appropriately licensed NHS guidance and NICE material only where AI reuse is authorised.
- Return evidence snippets and traceable citations; generate clear plain-English explanations.
- Handle insufficient evidence and avoid unsupported medical assertions.

Deliverable: retrieve_evidence() and generate_explanation() with validated citations.

Member 3 | Frontend, access and sustainability

- Build upload, redacted-preview confirmation, explanation, follow-up and deletion confirmation screens.
- Prioritise mobile layout, keyboard access, readable language and transparent status messages.
- Record baseline response time, model size and memory/resource use where feasible.

Deliverable: Usable end-to-end interface connected to agreed backend contracts.

Member 4 | Research, fairness evaluation and presentation

- Create an illustrative patient persona and detailed user journey.
- Prepare 15-20 synthetic test cases covering fairness, safety, failure, privacy and access.
- Check source-use permissions; maintain evidence register, threat model, pitch and demo script.

Deliverable: Evaluation matrix, sources register, journey map and presentation assets.

2. Proposed patient journey and architecture

Consent and secure intake → OCR and Presidio redaction → Redacted image and values shown to user → Explicit approval → Trusted evidence retrieval → Bounded explanation with citations → Follow-up questions (optional) → End session and verified deletion.

Approval is a hard gate. The explanation module receives only the confirmed, redacted structured report; it never receives the original image. If the report is rejected, offer correction or deletion. If sources are missing or the request is outside scope, the system abstains or recommends qualified human help.

3. Tools and integration contracts

Operation	Owner	What it returns
extract_report()	1	Test values, units, ranges, OCR confidence

Operation	Owner	What it returns
redact_report()	1	Redacted image, removed identifier categories
confirm_report()	1 + 3	User decision and validated structured fields
retrieve_evidence()	2	Evidence passages, URLs, source IDs
generate_explanation()	2	Plain-English explanation with source references
validate_response()	1 + 2	Policy checks, citation checks, blocked claims
delete_session()	1	Deletion status for each retained artefact

4. Suggested stack

Backend: Python and FastAPI. **Orchestration:** LangGraph with deterministic human approval gates. **Extraction:** Tesseract/Pillow. **Privacy:** Presidio Analyzer, Anonymizer and Image Redactor. **Retrieval:** Sentence Transformers with FAISS or Chroma. **Local generation:** Ollama small instruct model. **Frontend:** Streamlit for speed, or existing Next.js scaffold. **Testing:** Pytest plus synthetic fixtures.

Use typed Python tools called by LangGraph nodes. MCP is optional and should not be introduced unless a concrete integration needs it. Avoid unconstrained autonomous agents or any tool that can prescribe medication.

5. Responsible AI ownership

Privacy and security: Member 1 builds redaction, least-privilege access, minimal retention and verified deletion; Member 4 tests it.

Explainability and trust: Member 2 retrieves approved sources, verifies citations and labels uncertainty.

Access: Member 3 builds simple wording, keyboard-accessible and mobile-friendly interfaces.

Bias and fairness: Member 4 checks equivalent synthetic tests with varied names and wording; Member 2 investigates inconsistent explanations.

Sustainability: Members 2 and 3 benchmark a small local model, cache reusable public evidence and report measured resource use.

6. Build order and demo exit criteria

- First hour: lock scope, choose one synthetic full blood count report, agree on JSON contracts and licensing constraints.
- Next three hours: build redaction, retrieval and UI in parallel using mocks.
- Following two hours: integrate upload, approval, grounded explanation and deletion in LangGraph.
- Remaining build time: add safety checks, access improvements, fairness tests and resource measurements.
- Before Sunday at 11 AM: freeze features, run tests, verify privacy and rehearse a complete patient journey.

Demo must prove: upload and understanding; visible redaction; user approval; grounded explanation; refusal of prescriptions; and verified data deletion. Use synthetic reports only. No claim of clinical validation.

Critical safety caution: Do not store raw images or identifying OCR output in LangGraph checkpoints, logs or analytics. Deletion must cover files, temporary artefacts, derived session state and checkpoints, and success must only be shown once verified.